



Linguistic, Orthographic, and Typographical Errors in Science (LOTS): A Growing Threat to Academic Integrity

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Abstract

Biomedical engineers produce knowledge and artifacts. Across the life cycle of an idea, errors can creep in. In this letter, we propose the term linguistic, orthographic, and typographical errors in science (LOTS) to represent a category of errors that threaten the truthfulness and integrity of scientific literature and engineering projects. They include documented cases of the misuse of generative artificial intelligence (GAI). LOTS consist of four categories: (1) simple spelling errors; (2) the semantic deformation of technical terms, in the form of ‘tortured phrases’; (3) letter or symbol-switching; and (4) formatting errors that impact the veracity of knowledge, or distort the precision of scientific representation, such as the absence or overuse of capitalization, the incorrect use or absence of italicization, the failure to deanonymize information, cloned template text, or GAI-generated “hallucinations.” We introduce small analyses to assess the incidence of dopamine- β -hydroxylase, β -secretase, and β -adrenoreceptors (erroneous *Eszett* formats) supposedly representing dopamine- β -hydroxylase (DBH), β -secretase, and β -adrenoreceptors, respectively, in PubMed. We suggest that virtuous biomedical engineers should address LOTS. To improve the screening of LOTS, we argue in favor of a common framework for scholarly text integrity analysis.

Keywords Biomedical engineering ethics · Errors in science · Neuroscience · Research ethics · Precision · Truthfulness · Tortured phrases

Dear *Annals of Biomedical Engineering* Editors,

Biomedical engineers produce both knowledge and artifacts [1]. To improve the likelihood of the realization of knowledge-driven as well as practical aims, the scientific literature (including that related to biomedical engineering) must accurately represent the objects that it studies. This makes the publication of claims lacking reproducibility or validity both “epistemically and socially irresponsible” [2]. Scientific authors, peer reviewers, and editors must work together to promote truthful research [3]. Untruthful claims

do not satisfy the criterion of “repeatable efficacy,” and their rejection is what makes modern science epistemically robust and therefore of high social value [3, 4].

Following documented cases of researcher misconduct, over recent decades, there has been a growing awareness of the importance of the integrity of the scientific literature as part of the broader problem of scientific integrity [4–6], coupled with the common knowledge that research papers are rarely flawless [7]. We add to this growing body of knowledge by describing linguistic, orthographic, and typographical errors in science (LOTS), which represent a category of scientific errors that do not meet the criterion of repeatable efficacy and are thus not truthful. We encourage a situation of shared responsibility for the reduction of LOTS.

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Linguistic, Orthographic, and Typographical Errors in Science

Nylenna and Simonsen [6] applied a population-level perspective on preventing the erosion of scientific integrity and established a continuum of errors within science from error

to misconduct to full-blown fraud. Building on that framework, Bolland et al. [5] classified “typographical errors” as the lowest severity of errors in their taxonomy, compared to blatant misconduct such as data or figure manipulation. However, we note that within and around their category of typographical errors, there are distinct and related phenomena that we call LOTS, a growing category of heterogeneous errors that represent a threat to scientific integrity, particularly in the age of generative artificial intelligence (GAI). We identify four categories of LOTS: (1) simple spelling errors; (2) linguistic deformation, in the form of ‘tortured phrases’; (3) letter or symbol-switching; and (4) formatting errors that impact the veracity of knowledge, such as GAI-generated “hallucinations.”

Simple Spelling Errors

The introduction of spelling errors, colloquially referred to as “typos,” is likely human, i.e., generated by authors, but can also occur when there is no or poor editorial control prior to publication. Over 40 years ago, from about 25 million words of text that was archived in scientific databases, approximately 50,000 misspellings were detected [8]. Although it is impossible to understand precisely how such errors emerge in the scientific literature in the absence of any explanation by authors or editorial staff, like copy-editors, we envision four likely situations: (a) the lack of detection if an author or copy-editor is not a native English speaker; (b) older publications before the age of text processors and GAI; (c) the undetected replacement of one word with an incorrect substitute that is spelt similarly, e.g., six versus sex, but that would not be detected by spell-check software and could only be detectable following copy-editing of the words within the sentence’s context; and (d) authors or copy-editors

who do not employ spell check in their text processor, or the language that is activated in their text processor (e.g., Word) is not English, not allowing them to detect the spelling error. As a result, spelling errors are published undetected and remain uncorrected (select examples in Table 1). Even in the case of (b), it seems incredulous that biomedical journals would still contain spelling errors, considering that AI (natural language processing) tools already existed around the turn of the millennium [9] and have only strengthened since in terms of detection and correction [10].

Linguistic Deformation, as ‘Tortured Phrases’

The creation of nonsense synonymized terms for existing jargon by humans or GAI is known as the production of ‘tortured phrases’ [11]. These terms can arise when online thesauruses or software that synonymizes text are used, without the author being aware that a standard term or jargon has been replaced with a term that would not generally be accepted or recognized by peers in that scientific domain. On occasion, ‘tortured phrases’ may appear when authors have attempted to circumvent the plagiarism control by a journal, and in such egregious cases, they serve as primers, allowing other ethical impasses or misconduct to be detected [12]. A wide swathe of biomedical literature is being impacted, including in indexed journals that claim peer review and copy-editing-related quality control [13]. Concerted efforts are being made by the author(s) of this letter to detect ‘tortured phrases’ across a wide range of disciplines, including the biomedical sciences (Table 2).

Table 1 Random examples of spelling errors in the English titles of indexed scientific literature

Paper’s DOI (year; journal; publisher)	Spelling error	Correct spelling
https://doi.org/10.1111/j.0954-6820.1928.tb03943.x 1928; <i>Acta Medica Scandinavica</i> ; Wiley	vitimin	Vitamin
https://doi.org/10.1038/220429a0 1968; <i>Nature</i> ; Springer Nature ^a	Bonk	Bank
https://doi.org/10.1002/chin.198445030 1984; <i>Chemischer Informationsdienst</i> ; Wiley	aequous	Aqueous
https://doi.org/10.1016/0044-8486(91)90007-T 1991; <i>Aquaculture</i> ; Elsevier	tolarances	Tolerance
https://doi.org/10.1016/S0167-5273(10)70187-X 2010; <i>International Journal of Cardiology</i> ; Elsevier	streptokinaz rezistance	Streptokinase resistance
https://doi.org/10.15623/ijret.2016.0509031 2016; <i>International Journal of Research in Engineering and Technology</i> ; eSAT Publishing House	cocking oil	Cooking oil

Identified randomly by the second author; more systematic analyses are ongoing

^aIn this case, the PDF file lists the word correctly, but the HTML version, as well as the metadata and “How to cite” metadata list a spelling error. Found at: <https://www.pubpeer.com/publications/F847B484B17A50BB8C800D16AE4C95#1>

Table 2 Literature by the author(s) related to the identification and assessment of ‘tortured phrases’ across a wide range of disciplines or topics

Year of publication	Discipline/topic	DOI ^b
2026	Blood–brain barrier	10.2174/0115672050460224260206052444
2025	COVID-19	10.1093/nutrit/nuaf233
2025 ^a	World Health Organization	10.20944/preprints202510.1478.v1
2025 ^a	Blood–brain barrier	10.2139/ssrn.5131445
2025	Nucleic acids	10.35845/abms.2025.2.474
2025	AI	10.1186/s43067-025-00219-8
2025	Huntington’s disease	10.1515/ohe-2025-0070
2025	Entomology	10.1002/nse2.70006
2024	Creutzfeldt–Jakob disease	10.5603/mrj.102790
2024	Sport	10.36950/2024.9ciss010
2024	Copy-editing	10.36591/SE-4703-22
2024	Breast cancer	10.5603/njo.101687
2024	Environment	10.22364/eeb.22.14
2024	Organic acids	10.1007/s42535-023-00772-5
2024	ChatGPT; nursing	10.1097/NNE.0000000000001290
2024	CRISPR	10.1089/crispr.2024.0005
2024	General (linguistic)	10.1016/j.ejrad.2023.111212
2023	COVID-19; epistemology	10.5195/pom.2023.164
2023	Parkinson’s disease	10.1007/s10072-023-06672-5
2023	Preprints	10.1080/03007995.2023.2201098
2023	Neurosciences	10.1016/j.neuri.2023.100127
2022	Materials, computer and engineering sciences	10.48130/PR-2022-0006
2022	Lithium-ion batteries	10.30538/psrp-easl2022.0087
2022	General (medical jargon)	10.19204/2022/TRTR2
2021	Heterosexuality of the carbon structure	10.1016/j.rinp.2021.104842

^aPreprint^bSome DOIs are not recognized by Crossref

Letter or Symbol-Switching

The confusion of letters or technical symbols can lead to miscalculations within a manuscript, such as “mislabeling a standard deviation as a standard error of the mean, and vice-versa” [5]. Errors can also concern an individual letter, for instance, in the case where the Greek letter *beta* (β) is confused with (or erroneously switched by) the German letter *Eszett* (β) [14]. Select examples are presented in Table 3.

This form of “letter-switching” may have a downstream bibliometric impact, for instance, on the presence of a technical term in a search engine, affecting the citability of a paper that contains “letter/symbol-switching.”

To advance readers’ appreciation of this phenomenon and its fine-scale impact, especially on the nomenclature of biomedical compounds, we supplement known “letter/symbol-switching” cases with three small PubMed-based analyses, for dopamine- β -hydroxylase, β -secretase, and β -adrenoreceptors (erroneous *Eszett* formats) supposedly

Table 3 Examples of studies recording cases where the Greek letter *beta* (β) has been erroneously represented by the German letter *Eszett* (β)

Correct term or compound	Incorrect derivative term	DOI
<i>Beta</i> -carotene	<i>Eszett</i> -carotene	10.1016/j.nut.2021.111398; 10.1016/j.jfca.2022.105047
Amyloid- <i>beta</i>	Amyloid- <i>eszett</i>	10.1016/j.msard.2022.104249; 10.1111/ejn.16058
<i>Beta</i> -catenin	<i>Eszett</i> -catenin	10.1007/s00210-024-02979-y; 10.1108/JD-08-2025-0222
<i>Beta</i> -glucuronidase	<i>Eszett</i> -glucuronidase	10.1007/s42535-022-00452-w; 10.1108/JD-08-2025-0222
<i>Beta</i> -interferon	<i>Eszett</i> -interferon	10.5603/mrj.107206

Conceptual revelation of this phenomenon/linguistic error [14]

representing dopamine- β -hydroxylase (DBH), β -secretase, and β -adrenoreceptors, respectively.

DBH

DBH/D β H is a key enzyme catalyzing the conversion of dopamine to norepinephrine, and an essential neurotransmitter in central and peripheral nervous systems, with a wealth of research having been published until this decade, including on the gene, *D β H* [15]. Clinically, DBH deficiency is a rare autosomal recessive disorder that presents as severe neurogenic orthostatic hypotension and can be treated with L-threo-3, 4-dihydroxyphenylserine (L-DOPS), a synthetic precursor of noradrenaline, to address the blood pressure and postural symptoms of DBH deficiency [16]. The *D β H* gene can also exist in different polymorphisms, which have been associated with pathologies in behavioral regulation and decision-making across a lifetime, including pediatric attention deficits [17], alcoholism [18], nicotine dependence [19], and schizophrenia [20]. Finally, *D β H* polymorphisms interact with other risk genes to increase the risk of Alzheimer's disease, the most common neurodegenerative disease [21]. Until May 21, 2025, a PubMed search returned 6484 results of papers on the subject of D β H [22].

Given the different diseases in which DBH functioning is implicated, the integrity of the growing literature on DBH is relevant to a large community of patients with neurological diseases, who, on a community-wide level, suffer from stigma and poor access to treatments [23]. There have also been recent scandals related to image manipulation in neuroscience, suggesting that the literature may be lacking in truthfulness [24]. Improving the translation of neuroscientific findings into accessible treatments is thus an urgent ethical priority, and guaranteeing that the neuroscience literature is free from errors is one way to improve its quality.

A search was completed at PubMed on May 21, 2025, for dopamine-*Eszett*-hydroxylase and dopamine- β -hydroxylase [25], yielding 0 and 17 results, respectively. This result is also important from the standpoint of PubMed's reliability, because the former (i.e., alphabetized term) results serve as a confirmation that the enzyme "dopamine-*Eszett*-hydroxylase" is non-existent. However, the latter result, "dopamine- β -hydroxylase" or the non-existent DEH/D β H, indicates that PubMed is not being able to differentiate *beta* from *Eszett* in select cases. Whether that error has been introduced by PubMed, or whether it has originated from the publisher is not clear without examining the source material.

The 17 cases, which were published between 1989 and 2023, were examined in greater detail (Table 4). The paper (<https://doi.org/10.1007/s42485-022-00088-z>) with the highest frequency of "dopamine-*Eszett*-hydroxylase" had 27 instances of the full or abbreviated (DBH) term, including in one figure. Except for that paper, and one more (<https://doi.org/10.1016/j.ejphar.2014.07.027>), which had eight

instances throughout the text, including in two figure legends, the remaining cases had very few (1–3) instances, reflecting the likely innocuous selection of the symbol of the *beta*-similar *Eszett*. In a few of the papers, other members of the "*Eszett*" family of errors were discovered, "amyloid-*Eszett*" and "*Eszett*-receptors," which were meant to represent A β (amyloid- β) and *beta*-receptors, respectively. Finally, there are four cases in Table 4, which are marked as false positives, in which "dopamine-*Eszett*-hydroxylase" was detected as such on PubMed, but not in the original publications. This suggests that at some point during the transfer or administration of bibliometric data at or by PubMed, the letter *beta* was somehow replaced by the letter *Eszett*. It has been advocated that to deal with these issues, as well as other integrity-related issues, including linguistic misrepresentations, as they have occurred with the "*Eszett*" family of errors, PubMed needs to take a more hands-on approach, rather than a disclaimer-based attitude [3].

β -Secretase

β -Secretase is one of the three proteinases associated with the formation of the A β peptide, the amyloid precursor protein that is central to the amyloid hypothesis and thus a target for potential treatments related to AD [26, 27]. A β -secretase that has been amply researched is β -secretase-1 (BACE1) [28]. There is thus interest in preserving the accuracy of the term " β -secretase," as well as ensuring its precise linguistic use in indexed scientific literature related to A β and AD.

On May, 21–26, 2025, using a similar search strategy employed for DBH, PubMed was searched for documents related to conventional β -secretase and β -secretase, yielding 10,883 and 12 results, respectively. Focus was placed on the latter cases of this linguistic error. In all 12 cases, which were papers published between 2010 and 2025, errors were recorded at PubMed, while in at least 1 case, no such errors appeared on the original journal's website (Table 5), i.e., a false positive, suggesting erroneous database management at PubMed, which needs to take a pro-active stance at correcting these errors [3], together with the publishers of these documents. In three cases, these papers had dozens of instances of " β -secretase," whereas the other cases only had one, or at most, a handful of this error. Several of the papers also contained the non-existent A β .

β -Adrenoreceptors

Beta-adrenoceptors or β -adrenergic receptors (β -ARs) are an important drug target, particularly of β -blockers, in the treatment of cardiovascular diseases, CVDs [29]. While it may seem obvious to state, there is philological interest in

Table 4 Instances of dopamine- β -hydroxylase (dopamine-*Eszett*-hydroxylase) in PubMed

PubMed URL	DOI	Locations and notes (instances)
https://pubmed.ncbi.nlm.nih.gov/36277464/	10.1007/s42485-022-00088-z	Title; abstract; main text; tables; figure legends; Fig. 1; also includes “amyloid- <i>Eszett</i> ” (28)
https://pubmed.ncbi.nlm.nih.gov/25058908/	10.1016/j.ejphar.2014.07.027	Title; abstract; keywords; abbreviations; Figs. 4 and 5 legends; references (8)
https://pubmed.ncbi.nlm.nih.gov/34827485/	10.3390/brainsci11111486	Abstract; references (2)
https://pubmed.ncbi.nlm.nih.gov/37106020/	10.1038/s41598-023-34074-2	Abstract; main text (2)
https://pubmed.ncbi.nlm.nih.gov/26803288/	10.1016/j.jash.2015.12.011	Abstract; highlights (HTML only) (3)
https://pubmed.ncbi.nlm.nih.gov/36306200/	10.1111/jne.13188	Abstract; main text; Fig. 6 legend (3)
https://pubmed.ncbi.nlm.nih.gov/30593840/	10.1016/j.bbi.2018.12.009	Keywords; main text; Fig. 2 legend (3)
https://pubmed.ncbi.nlm.nih.gov/26190515/	10.1111/ane.12462	Abstract; keywords; main text (3); also includes “ <i>Eszett</i> -receptors”
https://pubmed.ncbi.nlm.nih.gov/23903565/	10.1590/S0001-37652013005000041	Abstract (HTML version only); main text; Figs. 2 and 3 legends (3); also includes one generic “ <i>Eszett</i> ”
https://pubmed.ncbi.nlm.nih.gov/11094045/	10.1161/01.cir.102.22.2765	Abstract ^a
https://pubmed.ncbi.nlm.nih.gov/12689603/	10.1016/s0197-0186(03)00010-x	Abstract (1)
https://pubmed.ncbi.nlm.nih.gov/10402270/	10.1002/(SICI)1097-0029(19990701)46:1<24::AID-JEMT3>3.0.CO;2-E	Abstract (1)
https://pubmed.ncbi.nlm.nih.gov/9582409/	10.1007/s004410051081	Abstract; keywords; main text; Fig. 7 legend (4)
https://pubmed.ncbi.nlm.nih.gov/9740613/	10.1161/01.hyp.32.3.473	Abstract ^a
https://pubmed.ncbi.nlm.nih.gov/11139483/	10.1161/01.res.88.1.117	Abstract ^a
https://pubmed.ncbi.nlm.nih.gov/11158248/	10.1046/j.1471-4159.2001.00024.x	Abstract ^a
https://pubmed.ncbi.nlm.nih.gov/19210448/	10.1111/j.1365-2826.1989.tb00096.x	Abstract (1) ^b

Search date: May 21, 2025; listed in order of appearance at PubMed (searched based on relevance)

^aFalse positive, where *Eszett* was represented as “ss,” and only on the PubMed HTML version and not the HTML or PDF versions of the original, suggesting a coding error at or by PubMed. In this paper, we consider “ β ” to be the true positive.

^bRepresented as “ss” on PubMed HTML version and as *Eszett* on original page HTML version, but seemingly as beta in PDF (but text setting in old papers not crystal clear)

preserving the accuracy of the term β -AR, particularly in indexed medical literature.

Using a similar search strategy employed for DBH, nine papers were detected (May 21–26, 2025) at PubMed in which β -AR has been misrepresented as *Eszett* (β)-AR (Table 6). In five of those papers, the error appears only in the abstract and, in some instances, only at PubMed, suggesting that they are false positives. Conversely, only two papers had sizeable (11 and 54) instances of β -AR. Considering that only nine documents were discovered, against 39,415 papers with the correct representation of β -AR, i.e., that the issue is insignificantly small, and considering that other *Eszett*-related errors have been discovered in terms like A β rather than A β within a much larger body of indexed literature (see Table 3) than β -AR, is placing emphasis on this linguistic error making a scientific mountain from a linguistic mole-hill?

However, there may be wider associations, such as with β -blockers, and the misrepresentation of that term. If so, then the impact on CVD literature would be important to assess. Ultimately, it is up to CVD researchers to assess the impact of this error in their literature that they use and cite

to appreciate if there is value in assessing its impact on the scientific message, or whether this is merely “much ado about nothing.”

Summary and Limitations

The analyses in “DBH”, “ β -Secretase”, and “ β -Adrenoreceptors” sections were conducted purely from the perspective of a linguistic understanding of the “*Eszett*” family of errors, initially formalized in 2021 [14]. We in no way imply any intentional misconduct by the authors of the papers listed in Tables 4, 5, and 6. Rather, we use this opportunity to encourage authors, peer reviewers, and editors to check for cases where this error can arise and to only publish standard scientific terminology.

There is also the risk of making an error with older texts, in which the font and text setting, including in PDF files, is not as precise as it is in modern (around the early 2000s) files. This might even be the case for some of the cases in Tables 4, 5, and 6, including the false positives. Therefore, especially in texts where *beta* and *Eszett* may have been italicized (β and β , respectively), it is unclear whether publishers

Table 5 Instances of β -secretase (i.e., *Eszett*-secretase) in PubMed-indexed papers

PubMed URL DOI (date)	Locations in paper	Other cases of <i>Eszett</i>
https://pubmed.ncbi.nlm.nih.gov/40184913/ 10.1016/j.cyto.2025.156925 (2025) ^a	Abstract; main text	
https://pubmed.ncbi.nlm.nih.gov/39738076/ 10.1038/s41538-024-00360-0 (2024)	Abstract; main text	“amyloid- <i>Eszett</i> ”
https://pubmed.ncbi.nlm.nih.gov/36873999/ 10.3389/fphar.2023.1103012 (2023)	Abstract; main text ^b	“amyloid- <i>Eszett</i> ”
https://pubmed.ncbi.nlm.nih.gov/3598821/ 10.1016/j.mad.2022.111726 (2022)	Title; abstract; keywords; main text	“amyloid- <i>Eszett</i> ,” “ <i>Eszett</i> site,” “ <i>Eszett</i> - generated”
https://pubmed.ncbi.nlm.nih.gov/34267657/ 10.3389/fphar.2021.679335 (2021)	Title; abstract; main text ^c	
https://pubmed.ncbi.nlm.nih.gov/28841840/ 10.1186/s12883-017-0942-y (2017)	Abstract; abbreviations; main text	“amyloid- <i>Eszett</i> ”
https://pubmed.ncbi.nlm.nih.gov/27505013/ No DOI (2016)	Keywords; main text; table; references ^c	“amyloid- <i>Eszett</i> ”
https://pubmed.ncbi.nlm.nih.gov/26858599/ 10.3389/fnmol.2016.00004 (2016)	Abstract	
https://pubmed.ncbi.nlm.nih.gov/27022655/ 10.1016/j.biocel.2015.10.005 (2015)	Title ^d	
https://pubmed.ncbi.nlm.nih.gov/24404197/ 10.1371/journal.pone.0084962 (2014)	Abstract; main text	“amyloid- <i>Eszett</i> ”
https://pubmed.ncbi.nlm.nih.gov/22913824/ No DOI (2012)	Abstract ^{e,f}	“amyloid- <i>Eszett</i> ”
https://pubmed.ncbi.nlm.nih.gov/20675165/ 10.1016/j.ymgme.2010.07.007 (2010)	Title; main text; references ^c	“amyloid- <i>Eszett</i> ”

Search and analysis dates: May 21–26, 2025; listed chronologically, from newest to oldest

^aSubmission to this journal in May 2025 was met with a rejection in June 2025, with a promise by the Editor-in-Chief to issue a correction

^bIncidentally, the title contains an error: rather than “compound K,” “ompound K” has been published at PubMed, including in the citation, although this error does not appear in the HTML and PDF versions of the paper, as noticeable at the journal’s website: <https://www.frontiersin.org/journals/pharmacology/articles/10.3389/fphar.2023.1103012/full>

^cDozens of instances

^dThe error does not appear in the original title (<https://www.sciencedirect.com/science/article/abs/pii/S1357272515300327>), or anywhere in the original text, suggesting that the error was introduced by PubMed. In essence, this is a false positive

^eFull text was not available

^fThe PubMed citation also incorrectly lists the authors’ first names rather than the family names, so the citation has been corrected

and their text setters and copy-editors had an appreciation of this linguistic finesse, or even if they had access to text editors that would have been able to differentiate the two letters. A linguistic deep-dive into this phenomenon, which may very well have a historical explanation, would be appreciated by specialists.

Of note, searches at PubMed are not in the full texts, unless those are available in PubMedCentral and are mostly limited to the title, abstract, and keywords. Therefore, the cases in Tables 4, 5, and 6 are not a full complement of literature that might contain the terms dopamine- β -hydroxylase, β -secretase, and β -adrenoreceptors, and searches in large databases like Scopus or Web of Science, or even public search platforms like Google Scholar might reveal additional literature, including gray literature. For example, a

cursory search at Google Scholar on May 31, 2025, revealed 3300 results for “dopamine-*Eszett*-hydroxylase,” much more than in PubMed, although without a detailed analysis, it is impossible to appreciate the volume of true and negative results, an exercise that may have merit in future if DBH, β -secretase, and β -adrenoreceptors researchers decide to place emphasis on the “*Eszett*” family of errors.

Raising awareness, through these additional case studies, might allow AD and neuroscience researchers and editors to screen papers more meticulously for *Eszett*-related errors that are meant to represent *beta*-related words.

Table 6 Instances of β -adrenoreceptors (i.e., *Eszett*-adrenoreceptors) in PubMed-indexed documents

PubMed URL	Date; DOI	Locations (# instances); other notes
https://pubmed.ncbi.nlm.nih.gov/34572106/	2021; 10.3390/cells10092456	Title (1)
https://pubmed.ncbi.nlm.nih.gov/38858312/	2024; 10.1007/s43440-024-00605-5	Abstract (1)
https://pubmed.ncbi.nlm.nih.gov/35187102/	2022; 10.3389/fcvm.2021.777463	Title; main text (11)
https://pubmed.ncbi.nlm.nih.gov/33185366/	2021; 10.3906/sag-2007-234	Title; abstract; keywords; main text; references (54)
https://pubmed.ncbi.nlm.nih.gov/33002334/	2021; 10.1111/apha.13564	Keywords (2); also “ <i>Eszett</i> -AR-mediated inotropy”; “ <i>Eszett</i> -blocker”; “ <i>Eszett</i> -adrenergic”
https://pubmed.ncbi.nlm.nih.gov/11156881/	2001; 10.1161/01.cir.103.5.691	Abstract ^a
https://pubmed.ncbi.nlm.nih.gov/14998743/	2004; 10.1289/ehp.6755	Abstract ^b
https://pubmed.ncbi.nlm.nih.gov/10974331/	2000; 10.1016/s0028-3908(00)00088-5	Abstract ^c
https://pubmed.ncbi.nlm.nih.gov/25301123/	2014; no DOI	Abstract (1) ^d

Search and analysis dates: May 21–26, 2025; listed in order of appearance at PubMed

^a*Eszett* represented as “ss,” but only in the HTML version and only in the abstract, and only at PubMed, whereas in the PDF version, as well as the HTML abstract of the original journal page, the symbol beta is accurately represented

^b*Eszett* represented as “ss,” but only in the HTML version and only in the abstract, whereas in the PDF version, the symbol beta is accurately represented

^c*Eszett* represented as “ss,” but only in the HTML version and only in the abstract at PubMed, but as *Eszett* (symbol) in the abstract of the HTML and PDF versions of the Elsevier journal page

^d*Eszett* represented as “ss,” but only in the HTML version at PubMed and only in the abstract, whereas in the PDF version, the symbol beta is accurately represented

Formatting Errors

While scientists and authors of papers may generally associate the term “formatting” with a stylistic or visual branding of a journal, including the font type used, visual setting of information, or referencing style (AAPA, AMA, NML, etc.), the five types of formatting errors that we mention next impact the veracity of knowledge, or they distort the precision of scientific representation: the absence or overuse of capitalization; the incorrect use or absence of italicization; the failure to deanonymize information; cloned template text; and GAI-generated “hallucinations.”

The Absence or Overuse of Capitalization

Capital letters in the English language are traditionally assigned to proper nouns, and the non-standard or excessive use of capitals, including through error, can distort words that are purposefully supposed to either have a capital letter (e.g., an organism’s genus) or not (e.g., an organism’s species), even impacting the naming of genes, where a capital letter tends to represent a dominant gene and a lower-case letter refers to a mutant or recessive gene [30]. Capitalization can also be “abused” or purposefully “misused,” such as to emphasize social movements or causes, such as Open Access rather than open access, or to purposefully diminish entities [31], or petri dish rather than Petri dish [32]. Collectively, these single-letter errors can cause scientific and historical distortions within academic papers.

The Incorrect Use or Absence of Italicization

Italics have very specific uses in scientific writing, such as to represent the genus or species names of organisms like plants [33], or the names of genes, and conversely the absence of italics for transcription factors [34]. Consequently, the use of italics where italics is not required, or the absence of italics where it is necessary constitutes a minor scientific error. This type of error is particularly worrisome in indexed journals that claim peer review and that also presumably have copy-editing either prior to or during the proof stage. We offer two random examples¹ within plant science of the absence of italics in words that scientifically require it, or a “dual” stratum of use of italics, correctly in the main text, but incorrectly in the reference list. Consequently, the absence of italics in a text where it is also properly used could be considered a formatting “error,” or if it misrepresents the italic state due to a programming code error (e.g., in reference manager software), then it could also be considered a semantic error.

The Failure to Deanonymize Information

Some journals, in their “blinded” peer review format, request authors to remove information that could identify

¹ Example 1 (*PLoS ONE*): <https://doi.org/10.1371/journal.pone.0329099>; example 2 (*BMC Plant Science*): <https://doi.org/10.1186/s12870-023-04206-x>.

Table 7 The classification of AI “hallucinations,” leading to the falsification of truth

Recommended term ^a	Existing uses (verbatim quotes)
Confabulation	AI generated responses that sound plausible but are, in fact, incorrect
Delusion	AI generated responses that are false
Stochastic parroting	The repetition of training data or its patterns, rather than actual understanding or reasoning; LLM model generates confident, specific, and fluent answers that are factually completely wrong
Factual errors	Inaccuracies in information or statements that are not in accordance with reality or the truth, often unintentional but resulting in incorrect or misleading information
Fact fabrication	The occurrence where inaccurate information is invented, not represented in the training dataset, and is presented lucidly
Fabrication	The phenomenon where, as a generative AI, ChatGPT generates outputs based on statistical prediction of the text without human-like reasoning, potentially resulting in plausible-sounding but inaccurate responses; The phenomenon in ChatGPT output where the text is cogent but not necessarily true
Mistakes, blunders, false-hoods	Answers that are fabricated when data are insufficient for an accurate response
Hasty generalizations, false analogy, false dilemma	AI models making inferences that do not follow from the premises; also “hasty generalizations,” i.e., the fallacy of making (too) strong claims based on (too) limited data

Adopted and modified from the following source: <https://doi.org/10.48550/arXiv.2401.06796> (CC-BY license); preprint of Maleki et al. [38]

LLM large language model

^aSeveral of these categories could also fall into the categories of bias, hype, and spin

them, such as their names, affiliations, emails, and even funding sources. Those editorial actions are taken with a good intent in mind, namely to minimize bias and to try and improve the impartiality of the peer review. However, on occasion, papers that have passed an anonymization process fail to reinsert information that had been removed, and in such cases, the authors, copy-editors, and editors have not detected this missing information, leading to the publication of “XXX” in papers, or other text that represents the vestigial remnants of blinded peer review [35]. Those texts in essence carry textual errors in need of correction.

Cloned Template Text

Some journals employ template files that allow authors to insert their own paper’s information into allocated slots, such as the title, affiliations, abstract, and so on, thus allowing the paper to automatically follow the journal’s style, ultimately saving publishers time and resources in the case of an acceptance, or wasting authors’ time and resources if their paper is ultimately rejected from that journal. Nonetheless, such template files may carry established text, such as “Insert text here” or similar cloned instructions or other text that forms part of the template itself. On occasion, some papers are published with such cloned template text intact, i.e., that was not detected by the authors, copy-editors, and editors [36]. One such example is the vestigial remnants of a pangram “quick brown fox jumps over the lazy dog,” which journals (specifically editors and copy-editors) have failed to detect and forgotten to remove [37]. As for the other cases in this section, vestigial template text erodes the textual

integrity of published papers and should be corrected, whenever possible.

GAI-Generated “Hallucinations”

The output of GAI can be erroneous and has in broad terms been referred to as “hallucinations.” However, the term “hallucinations” has been used in a wide array of contexts, but all of which curiously fall into our LOTS classification scheme. We thus draw our classification of “hallucinations” as LOTS from the classification suggested by Maleki et al. [38] (Table 7).

Conclusion: Biomedical Engineers Should Address LOTS

If scientific publications are understood as tools that engineers can use to certain ends [39], then so-called linguistic, orthographic, and typographical errors in science (LOTS) contribute toward making the impacted scientific literature an unsafe work place by being sources of unreliable information [39]. We argue that biomedical engineers should address LOTS. We acknowledge that this rests on a “virtue ethics” approach to engineering [1]. That is, we defend a moral worldview of biomedical engineering that is not just about the field resolving technical problems in a passive way. We see biomedical engineering as a rich field full of both technical and normative problems, in which biomedical engineers should practice the virtues of thoroughness and care across the life cycle of an idea or product [1]. This includes an emphasis on reduction of errors that may

initially seem trivial, but which could have unforeseen consequences. However, irrespective of potential consequences, taking the time to correct errors also trains engineers to be rigorous and is thus intrinsically good [1]. From a virtue ethics perspective, engineers wishing to cultivate practical wisdom cannot pretend that “small” errors are unimportant. Deliberate intentions to deceive are particularly unvirtuous, but sloppiness—a lack of thorough attention to detail—should be corrected.

There is a robust set of corrective mechanisms to correct errors in the scientific literature [40]. Efforts to address LOTS could take place within a combined framework of scholarly text integrity analysis. Such a framework could use existing methods and emerging automated methods to flag instances and study the downstream impact of LOTS on citation integrity [41]. Countermeasures would include automated pre-submission tools (e.g., Unicode validators, ‘tortured phrase’ detectors) and policy reforms (that should include GAI disclosure [42] and the mandating of symbol standardization). We encourage the establishment of a common metric that journals could potentially use in the form of AI-assisted screening of submitted and published manuscripts by editors, for example, through the establishment of a Text Corruption Index.

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Declarations

Conflict of interest The authors declare no relevant conflicts of interest.

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